IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

  This project followed a complete data science workflow including **data collection, data wrangling, exploratory data analysis and interactive visualization** to learn with patterns in SpaceX launch data.
  SQL queries, maps with **Folium** and dashboards with **Plotly Dash** were used to analyze and visualize launch outcomes and site performance.
  Finally, **predictive modeling using Logistic Regression and GridSearchCV from scikit-learn** was used to predict rocket landing success and evaluate model accuracy.

- Summary of all results

  The analysis revealed patterns in launch success related to payload mass, orbit type, and launch site, helping identify factors that influence rocket landing outcomes. Interactive visualizations using **Folium** maps and dashboards built with **Plotly Dash** provided geographic and operational insights into launch activities of **SpaceX**. Finally, predictive models developed with **scikit-learn** achieved strong accuracy in predicting landing success, demonstrating the effectiveness of data-driven approaches for analyzing launch performance.

Introduction

The rapid growth of the commercial space industry has increased competition in satellite launch services. **SpaceX** has gained a competitive advantage by developing reusable rockets, significantly reducing the cost of space missions.

Understanding the factors that influence the **success of rocket landings** is essential for improving mission efficiency and reducing operational costs. This project analyzes historical launch data to identify patterns, visualize launch activities, and build predictive models capable of estimating the probability of landing success.

Section 1

Methodology

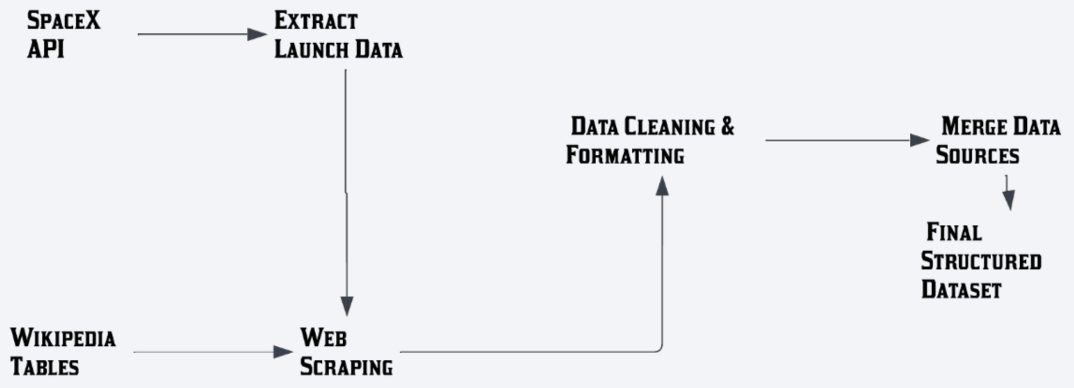
Methodology

Executive Summary

- Data collection methodology:
  - Describe how data was collected
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection

- Launch data was collected from two main sources: the **SpaceX API** and **Wikipedia tables**. The API provided structured information about launches, including rocket details, payload mass, orbit type, and landing outcomes, while Wikipedia was used to gather additional historical launch records through web scraping. After extraction, the data went through **cleaning and formatting**, where missing values and inconsistencies were handled. Finally, the datasets were **merged and organized into a structured dataset** using Python, preparing the data for exploratory analysis and predictive modeling.

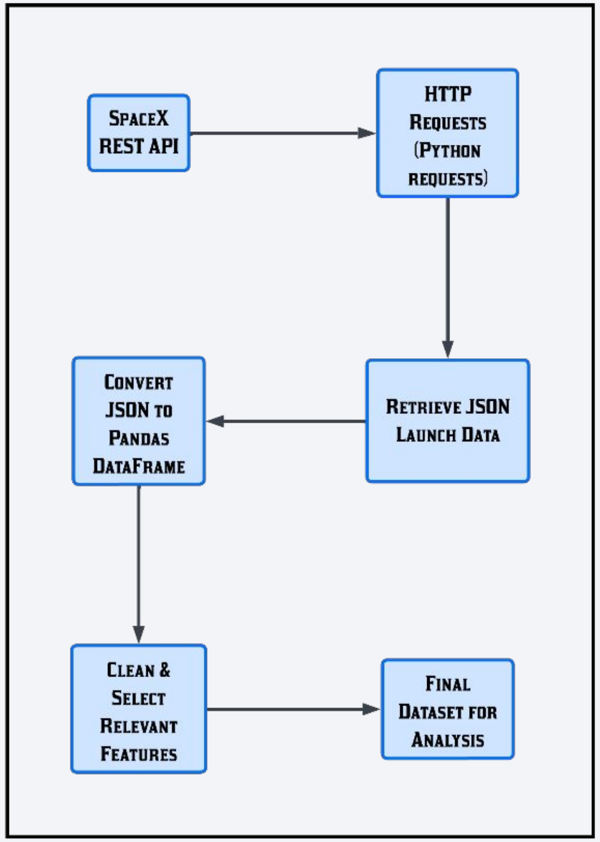


Data Collection – SpaceX API

- Launch data was collected through REST API calls to the **SpaceX API**, retrieving JSON responses containing detailed information about each mission. The data was processed using Python, converted into a pandas DataFrame, cleaned, and structured to create a dataset ready for exploratory analysis and predictive modeling.

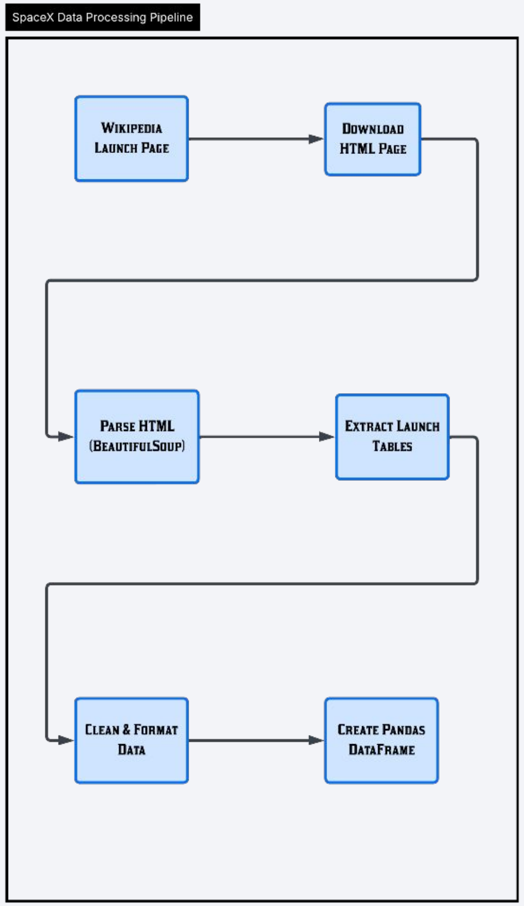
- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose

SpaceX Data Processing Pipeline



Data Collection - Scraping

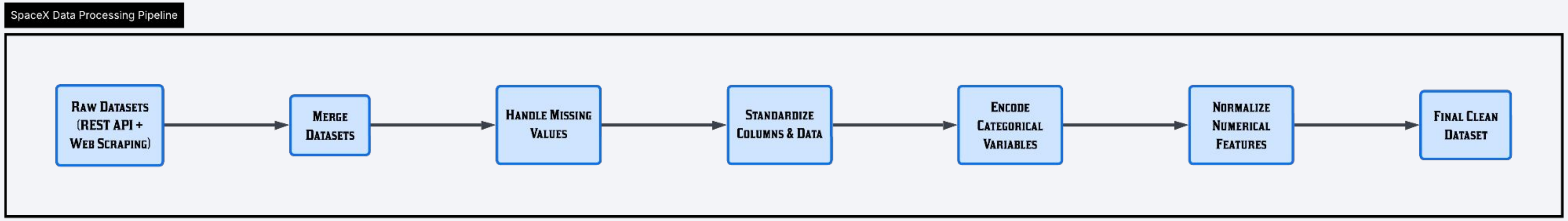
- Historical launch data was collected from Wikipedia using web scraping techniques. The HTML page was downloaded and parsed using Python and BeautifulSoup to extract launch tables. The extracted data was then cleaned, formatted, and converted into a structured pandas DataFrame for further analysis and visualization.



Data Wrangling

- Merged SpaceX API and Wikipedia datasets, handled **missing values** and standardized columns.
- Applied **One-Hot Encoding** to categorical features and **StandardScaler** to numerical features.
- Created a **clean, structured dataset** ready for EDA, visualization, and predictive modeling.

- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose



EDA with Data Visualization

- **Line Plots and Scatter Plots** - analyzed distributions and relationships between payload, orbit, and landing success.

- **Bar Charts** - compared success/failure rates across launch sites, orbits, and boosters.

Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose

EDA with SQL

- Several **SQL queries were performed to explore and analyze the SpaceX launch dataset**. The tasks included identifying **unique launch sites**, retrieving **specific launch records**, and calculating **metrics such as the total and average payload mass carried by different boosters**. The analysis also examined **landing outcomes**, including finding the **first successful ground pad landing** and counting **missions with successful and unsuccessful landings**. Additionally, more advanced queries were used to **identify boosters that carried specific payload ranges**, **determine the booster that carried the maximum payload using subqueries**, **analyze landing failures by month in 2015**, and **rank landing outcomes within a specific time period**.
- Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose

Build an Interactive Map with Folium

- Markers and circles were added to represent the **launch sites and launch events**, making their geographic locations easy to identify. Marker colors were used to distinguish **successful and failed landings**. Marker clusters were included to group multiple launches and keep the map readable. Lines were drawn to show the **distance between launch sites and nearby features** (such as cities, coastlines, or infrastructure), helping analyze the geographic context of the launch locations.
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

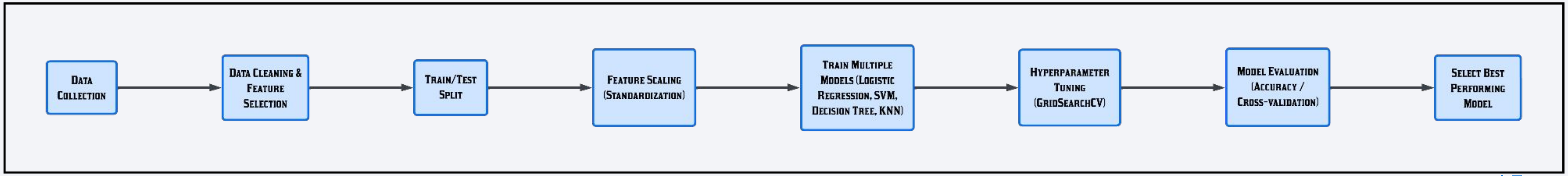
Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

- The model development process started with **data preparation**, which included cleaning the dataset, selecting relevant features, and splitting the data into **training and testing sets**. Next, **feature scaling** was applied to standardize the variables and improve model performance. Several classification models were then trained, including **Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)**. To improve performance, **hyperparameter tuning** was performed using **GridSearchCV** to identify the optimal parameters for each model. The models were then **evaluated using accuracy scores**, and the results were compared to determine the best-performing model. Finally, the **model with the highest performance** was selected as the final classification model.

SpaceX Data Processing Pipeline



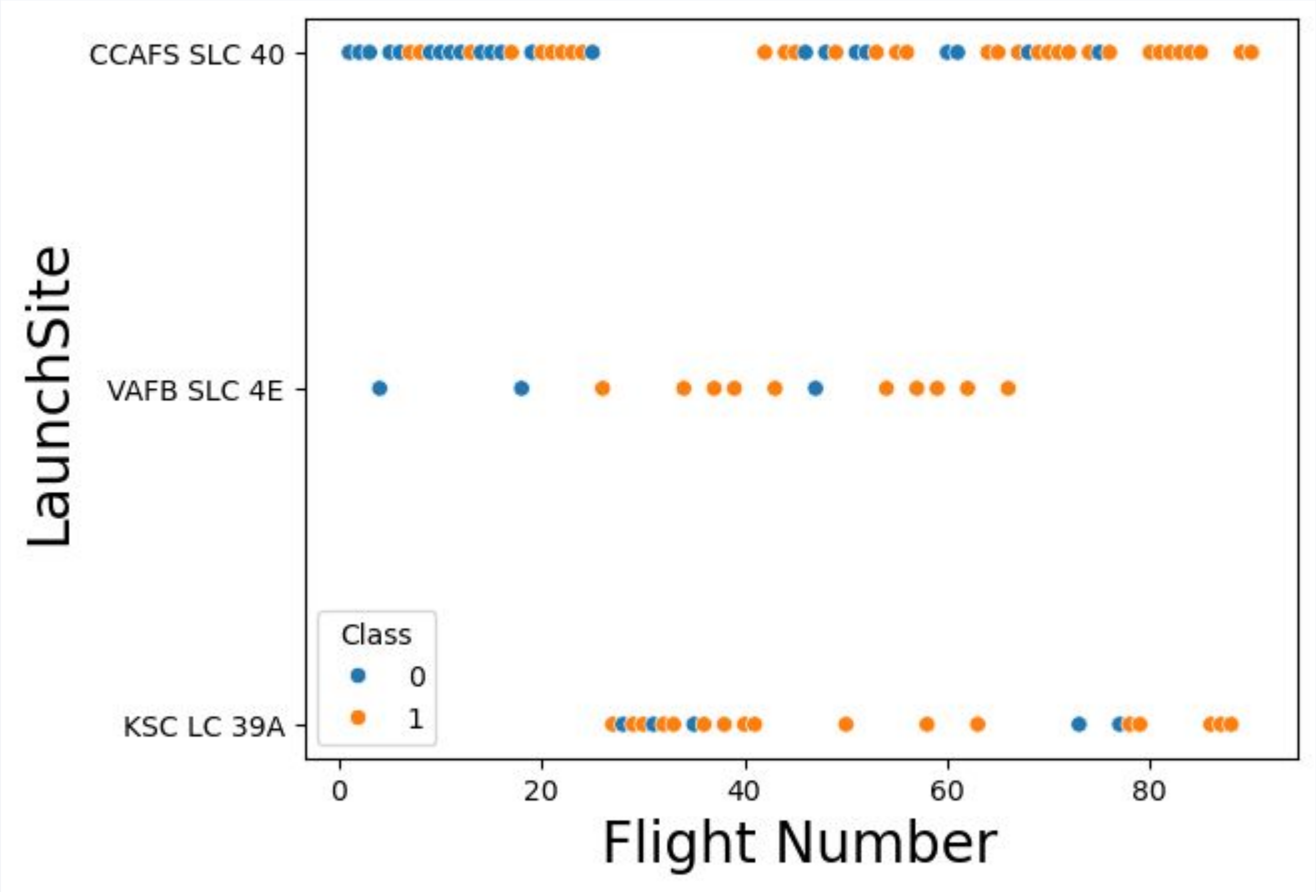
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

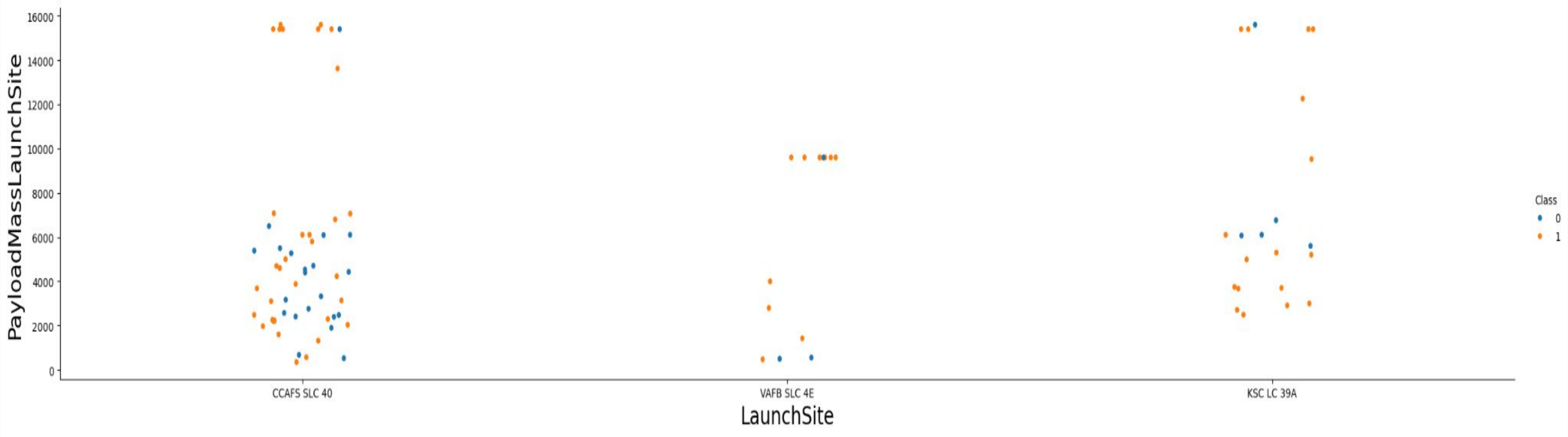
Section 2

Insights drawn from EDA

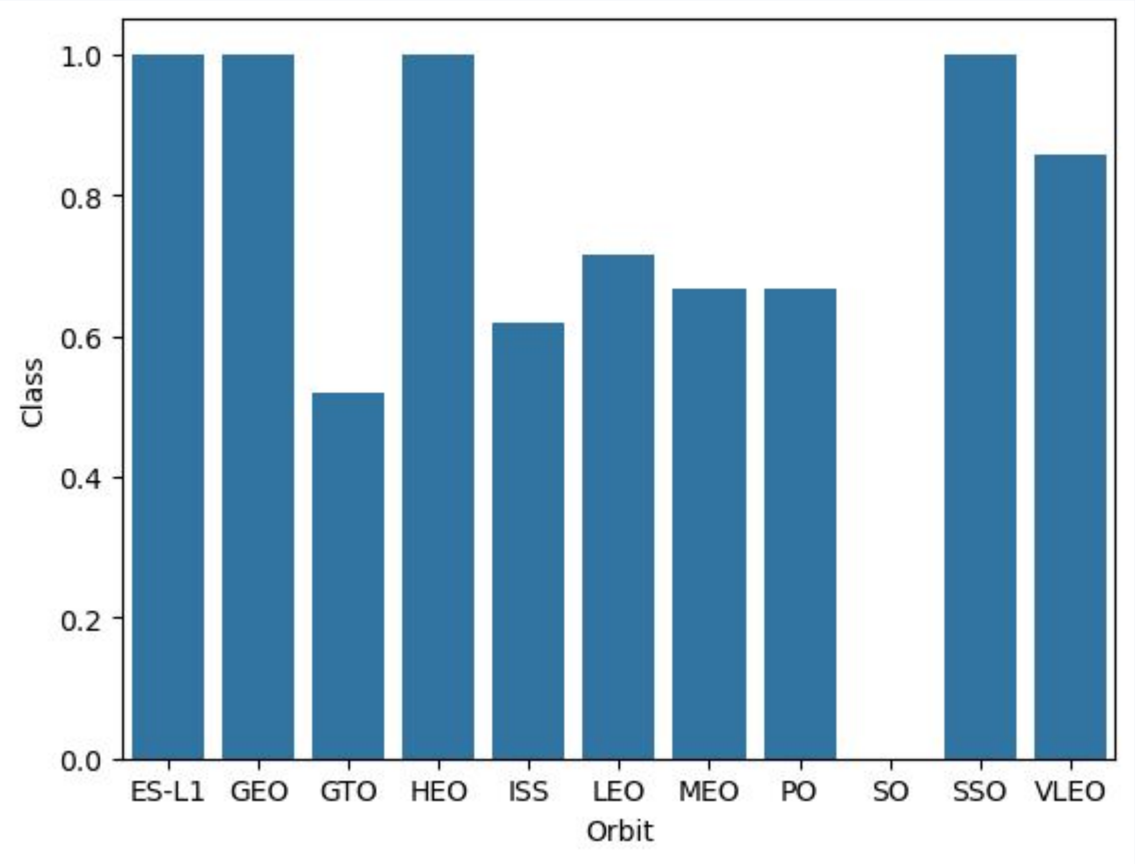
Flight Number vs. Launch Site



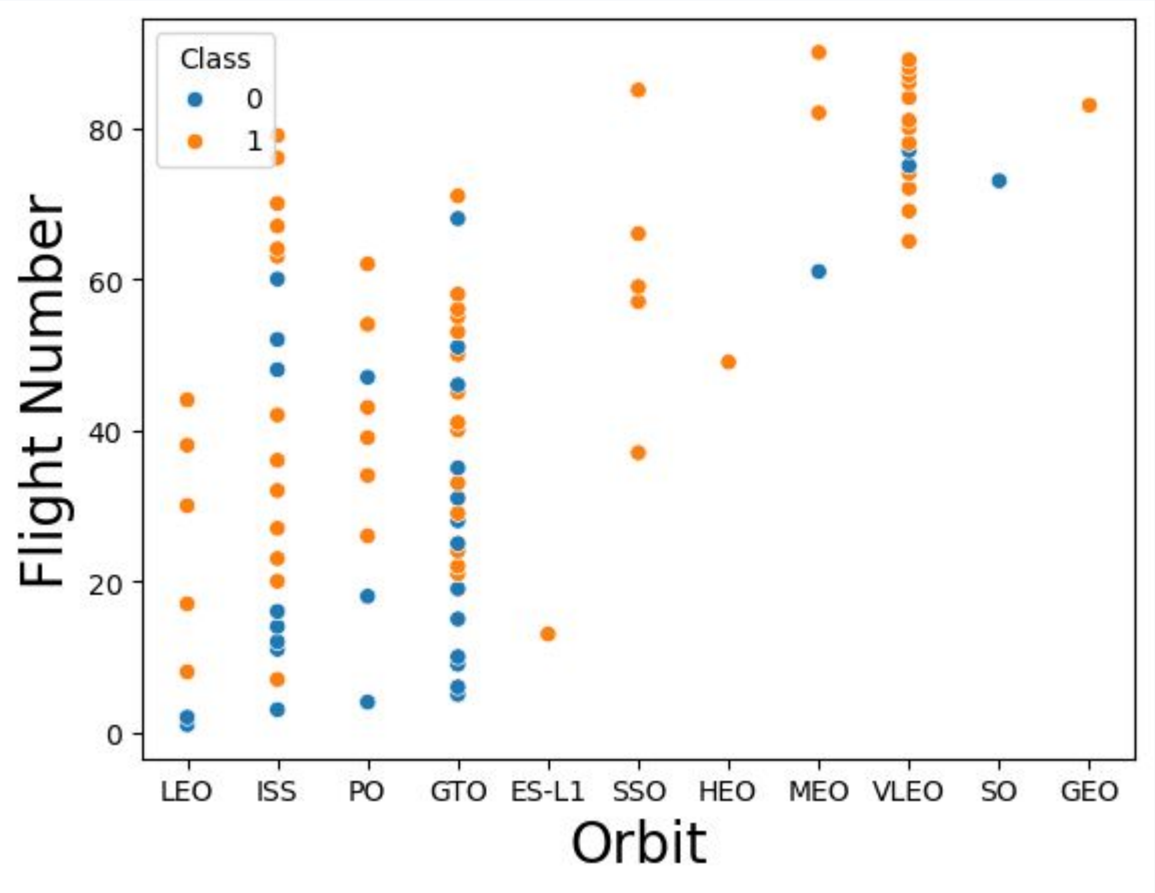
Payload vs. Launch Site



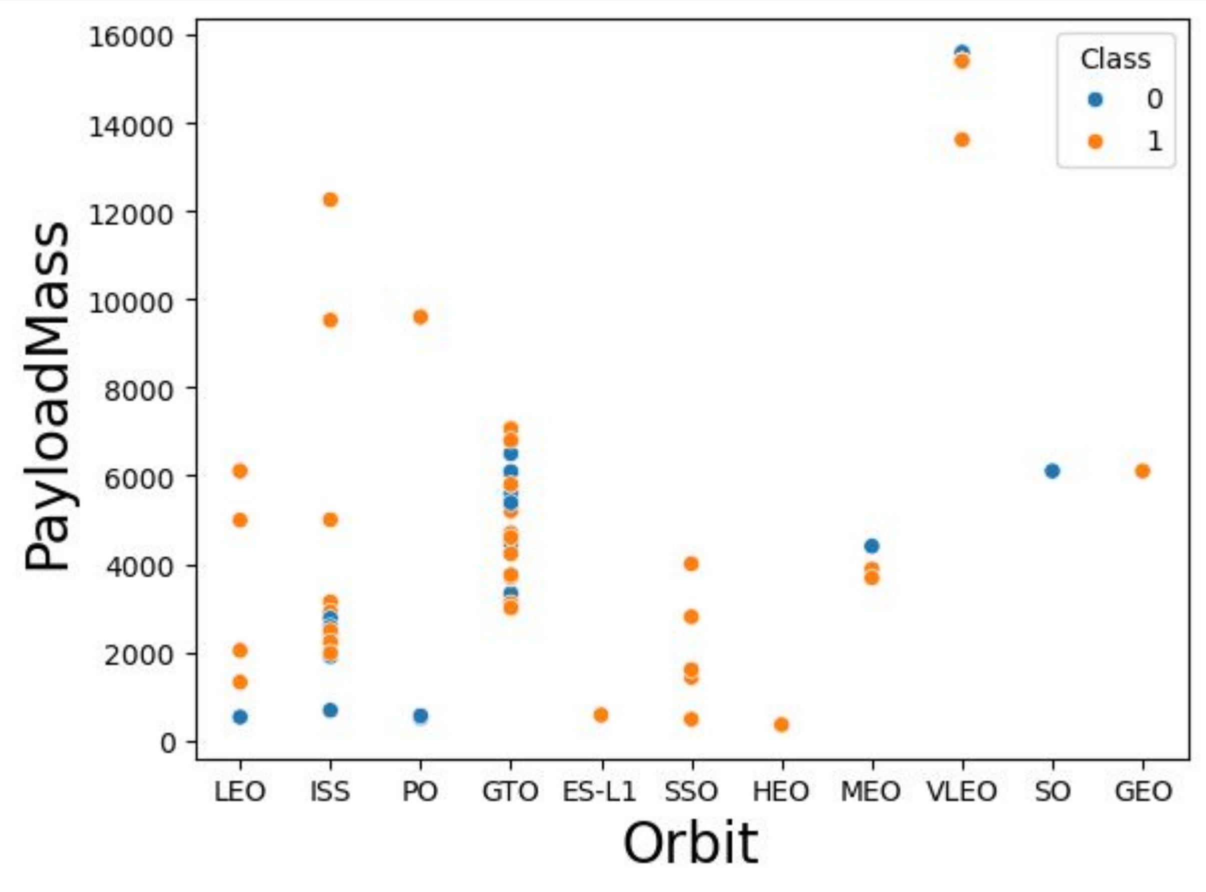
Success Rate vs. Orbit Type



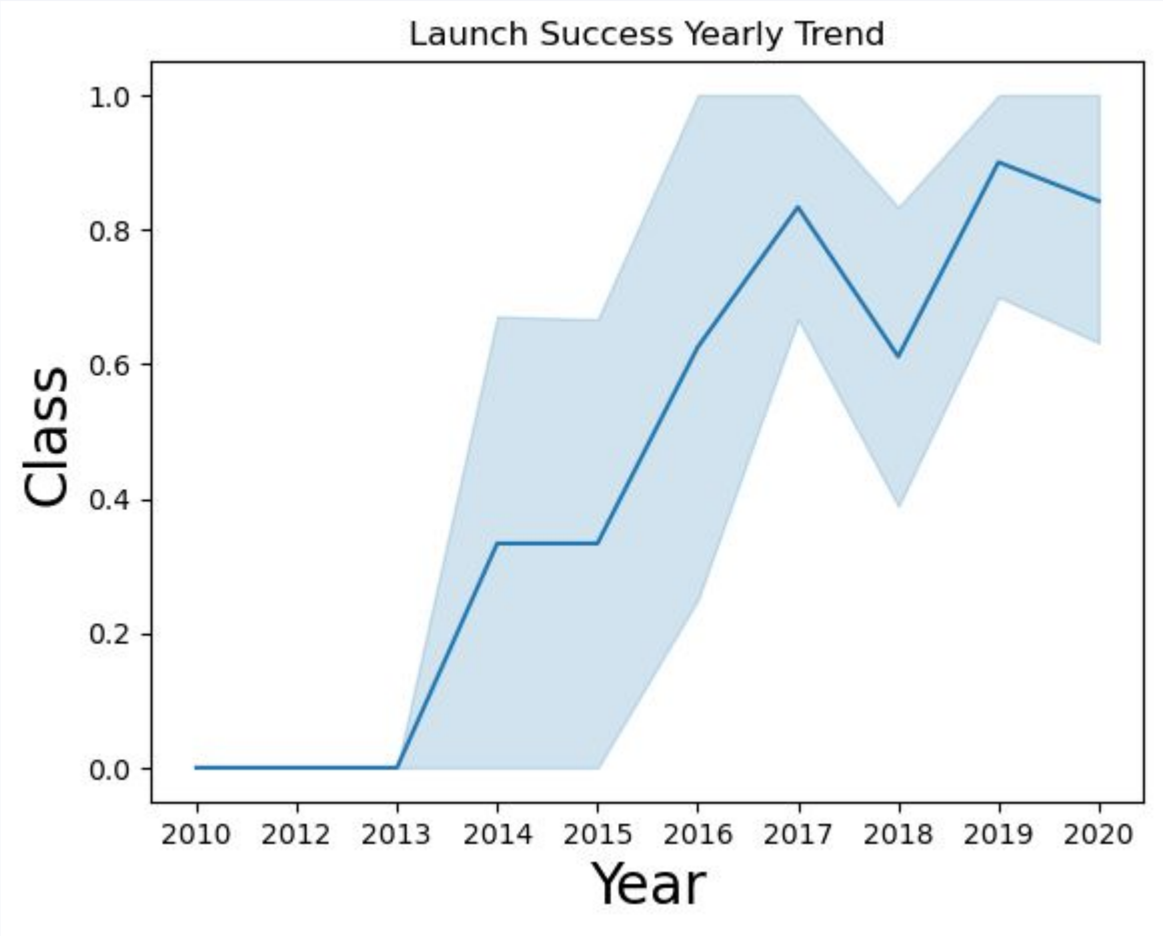
Flight Number vs. Orbit Type



Payload vs. Orbit Type



Launch Success Yearly Trend



All Launch Site Names

Query

```
SELECT DISTINCT(Launch_Site) FROM SPACEXTBL
```

Explanation

.Lists all unique launch sites in the dataset.

Results

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Query

SELECT * FROM SPACEXTBL WHERE Launch_Site like 'CCA%' limit 5

Explanation

Shows the first 5 launches from sites starting with "CCA".

Results

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Query

```
SELECT sum(PAYLOAD_MASS__KG_) AS Mass FROM SPACEXTBL WHERE  Customer = 'NASA (CRS)'
```

Explanation

Calculates the total payload mass for all launches by NASA (CRS).

Results

Mass
45596

Average Payload Mass by F9 v1.1

Query

```
SELECT sum(PAYLOAD_MASS__KG_) AS Mass FROM SPACEXTBL WHERE  Customer = 'NASA (CRS)'
```

Explanation

Calculates the **total payload mass** for all SpaceX launches contracted by NASA (CRS).

Results

MEAN
2928.4

First Successful Ground Landing Date

Query

```
select min(Date) AS DATE from SPACEXTBL where Landing_Outcome='Success (ground pad)'
```

Explanation

Finds the **earliest date** of a successful ground pad landing.

Results

DATE
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Query

```
select distinct(Booster_Version) from SPACEXTBL where Landing_Outcome='Success (drone ship)' and 4000<PAYLOAD_MASS__KG_<6000
```

Explanation

Returns unique booster versions that successfully landed on a drone ship with payloads between 4000 kg and 6000 kg.

Results

Booster_Version
F9 FT B1021.1
F9 FT B1022
F9 FT B1023.1
F9 FT B1026
F9 FT B1029.1
F9 FT B1021.2
F9 FT B1029.2
F9 FT B1036.1
F9 FT B1038.1
F9 B4 B1041.1
F9 FT B1031.2
F9 B4 B1042.1
F9 B4 B1045.1
F9 B5 B1046.1

Total Number of Successful and Failure Mission Outcomes

Query

```
%sql SELECT SUM(CASE WHEN Mission_Outcome = 'Success' THEN 1 ELSE 0 END) AS Success,SUM(CASE WHEN Mission_Outcome !=
'Success' THEN 1 ELSE 0 END) AS Fail FROM SPACEXTBL;
```

Explanation

Counts all successful and failed missions in a single row using conditional sums.

Results

Success	Fail
98	3

Boosters Carried Maximum Payload

Query

```
select distinct(Booster_Version) from SPACEXTBL where PAYLOAD_MASS__KG_= (select max(PAYLOAD_MASS__KG_) from SPACEXTBL
```

Explanation

Returns the booster version(s) that carried the heaviest payload.

Results

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Query

```
select distinct(Landing_Outcome) from SPACEXTBL
```

Explanation

Lists all unique landing outcomes in the dataset.

Results

Landing_Outcome
Failure (parachute)
No attempt
Uncontrolled (ocean)
Controlled (ocean)
Failure (drone ship)
Precluded (drone ship)
Success (ground pad)
Success (drone ship)
Success
Failure
No attempt

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query

```
select substr(Date, 6,2) as Month, Landing_Outcome,Booster_Version,launch_site  from SPACEXTBL where
substr(Date,0,5)='2015' and Landing_Outcome like 'Failure (drone ship)'
```

Explanation

Shows the month, landing outcome, booster version, and launch site for all 2015 launches that **failed on a drone ship**.

Results

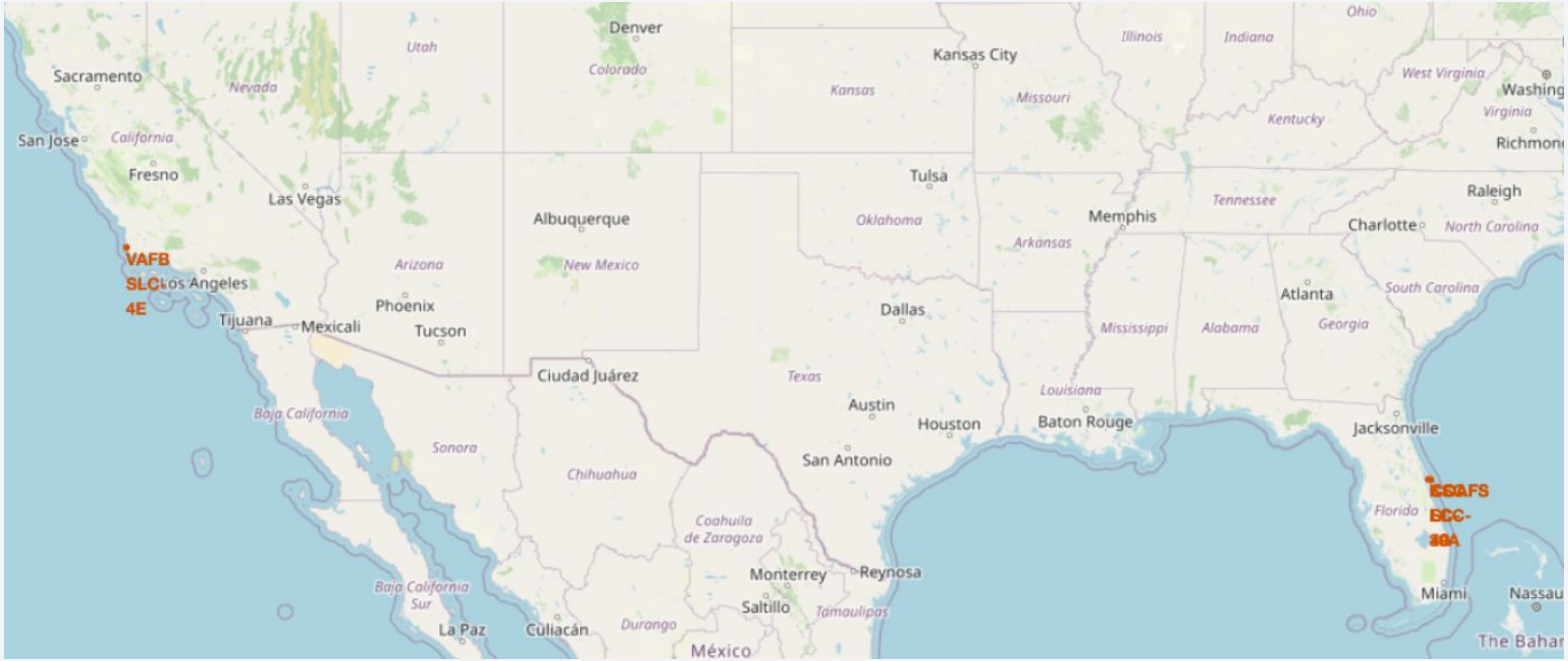
Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Section 3

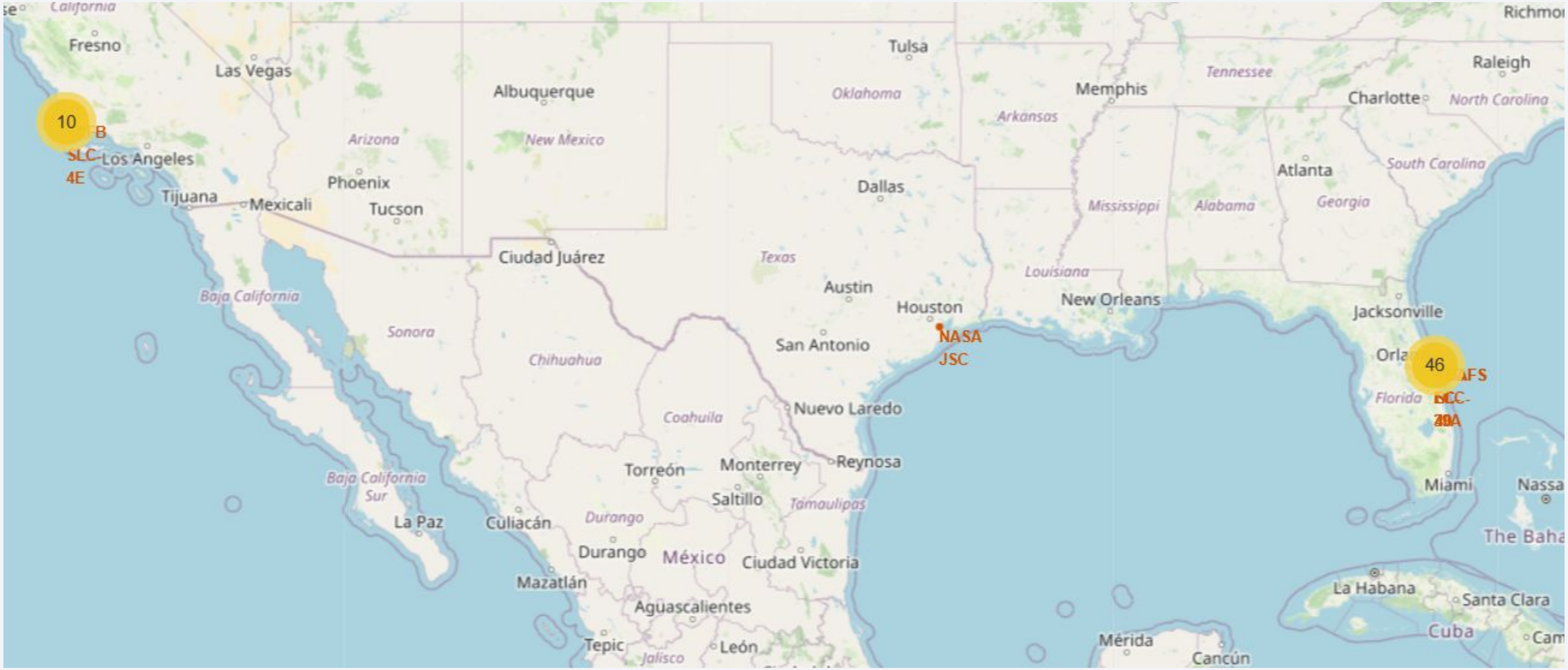
Launch Sites Proximities Analysis

Global SpaceX Launch Sites

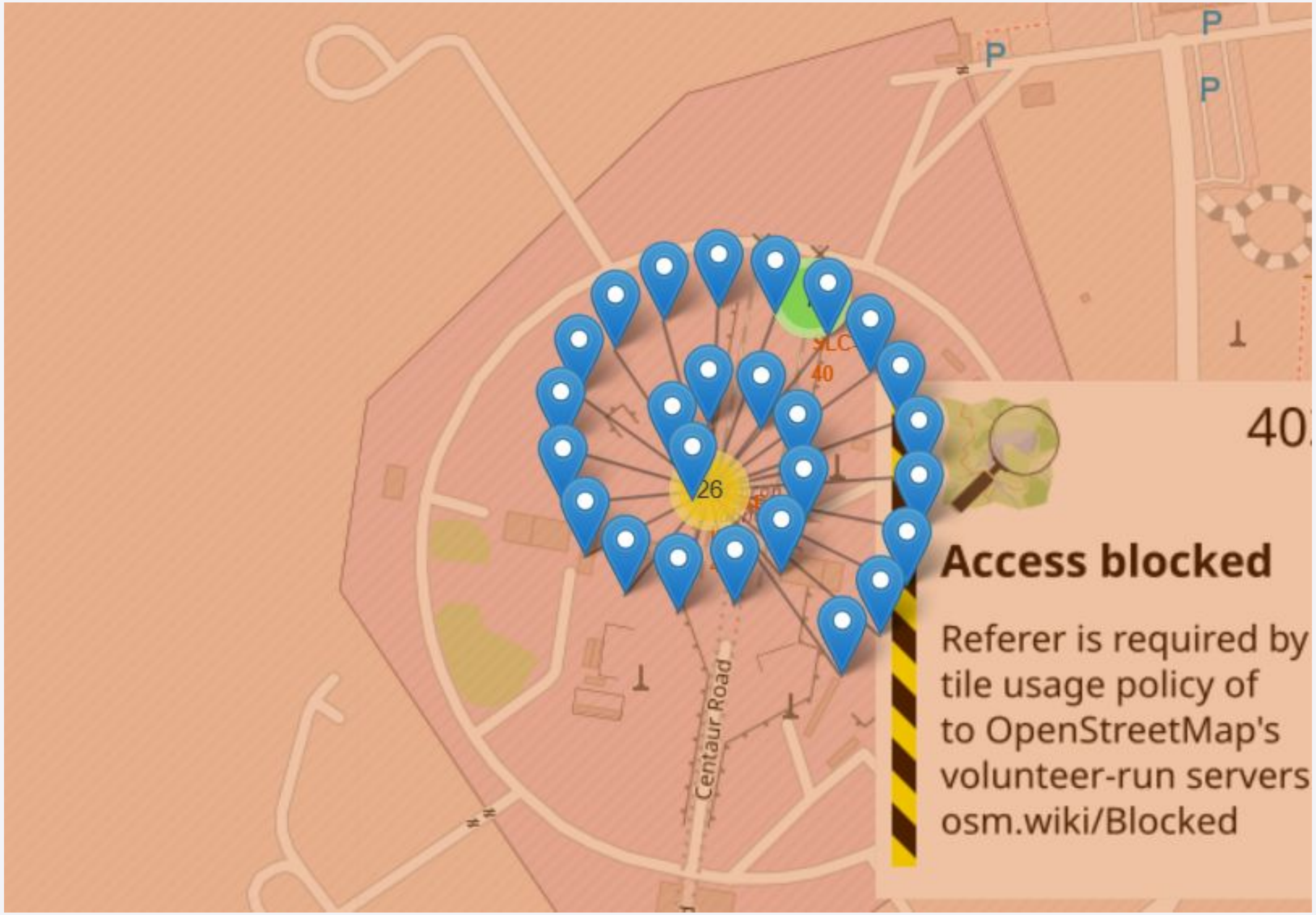
Every orange point is one launch site



Success/failed launches



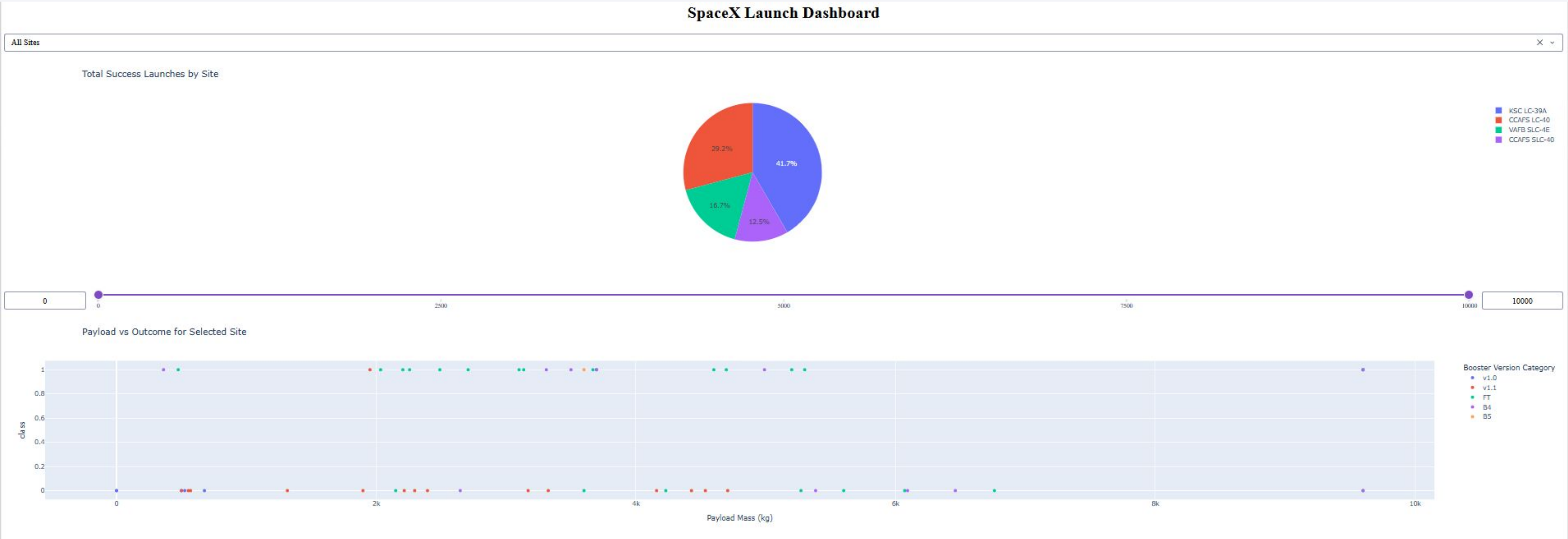
Success/failed launches +++ZOOM



Section 4

Build a Dashboard with Plotly Dash

Launch success count for all sites



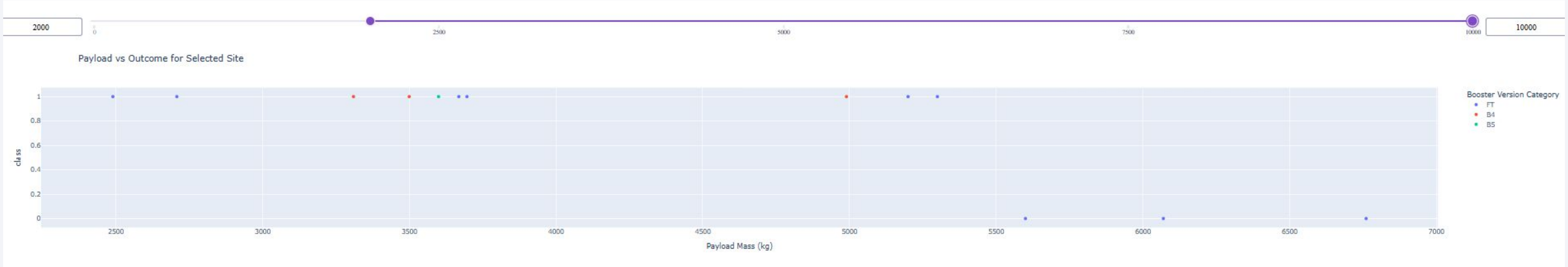
Launch site with highest launch success ratio

- Replace <Dashboard screenshot 2> title with an appropriate title

- Show the screenshot of the piechart for the launch site with highest launch success ratio

- Explain the important elements and findings on the screenshot

Payload vs. Launch Outcome scatter plot for all sites

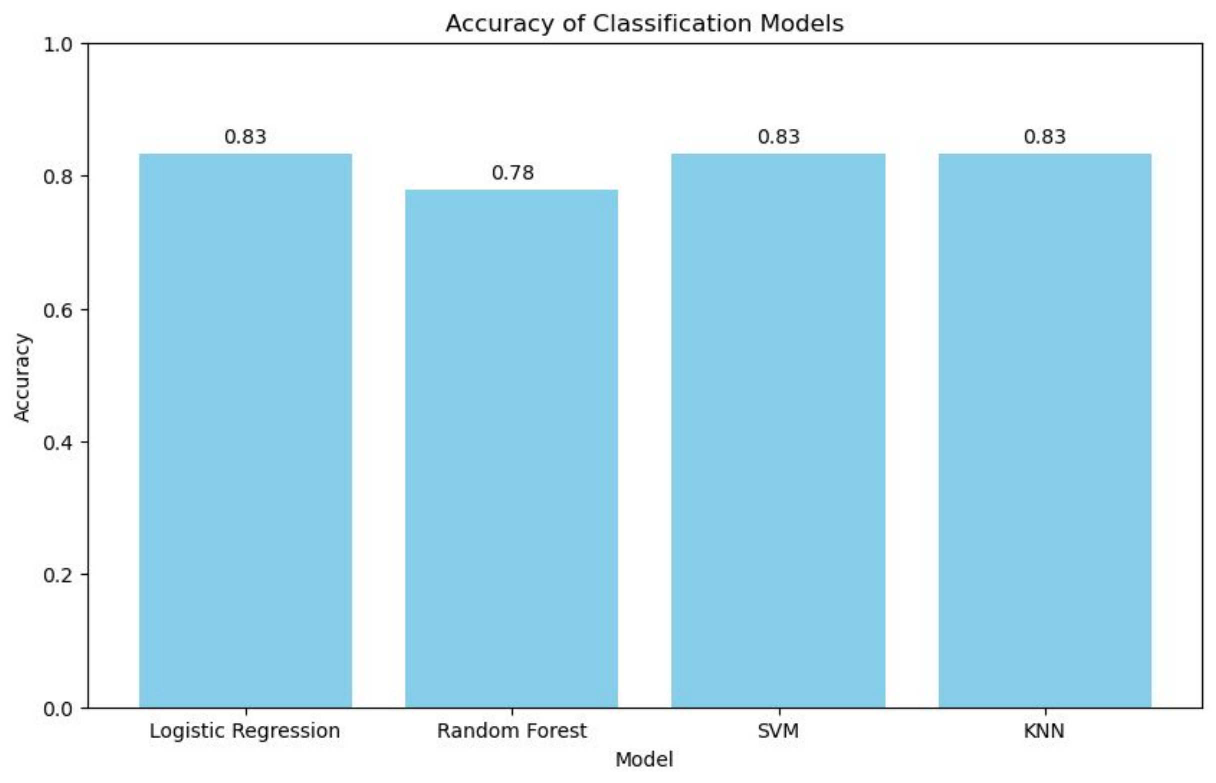


Section 5

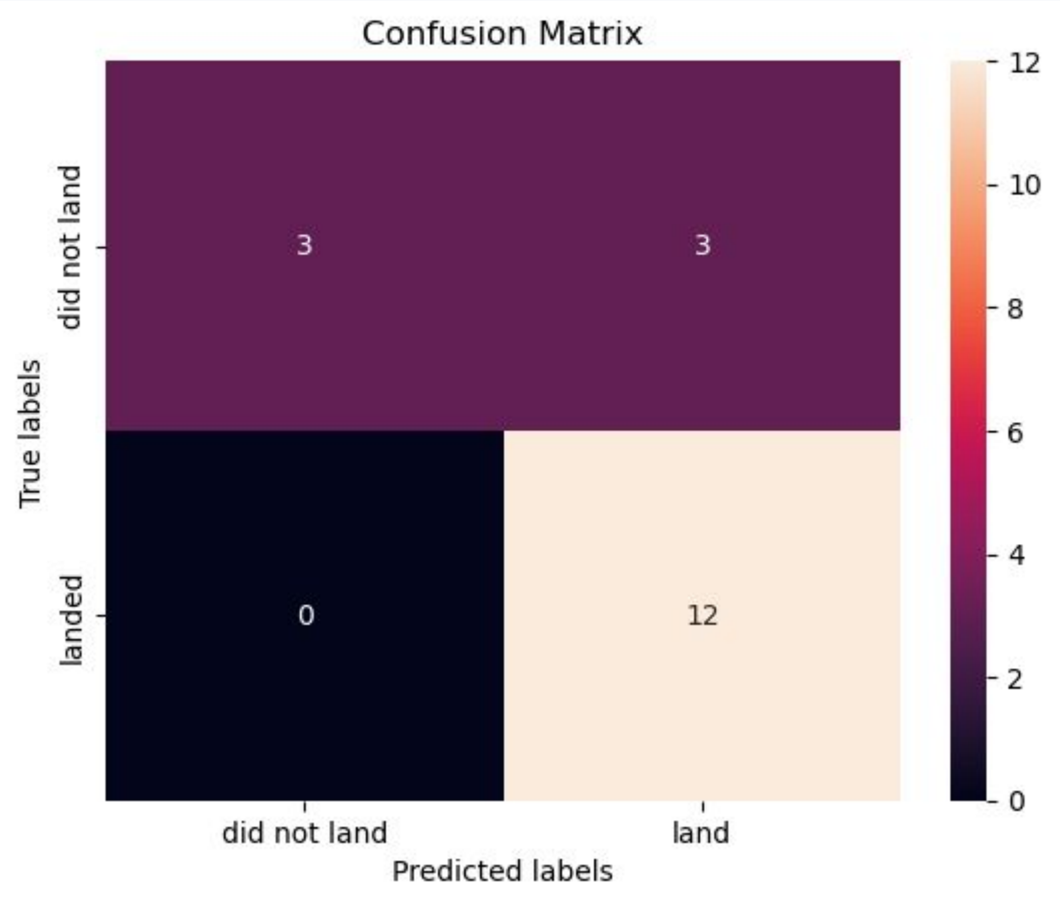
Predictive Analysis (Classification)

Classification Accuracy

- 3 of them have the same better accuracy so i choose the best accuracy in terms of training sets.



Confusion Matrix of KNN



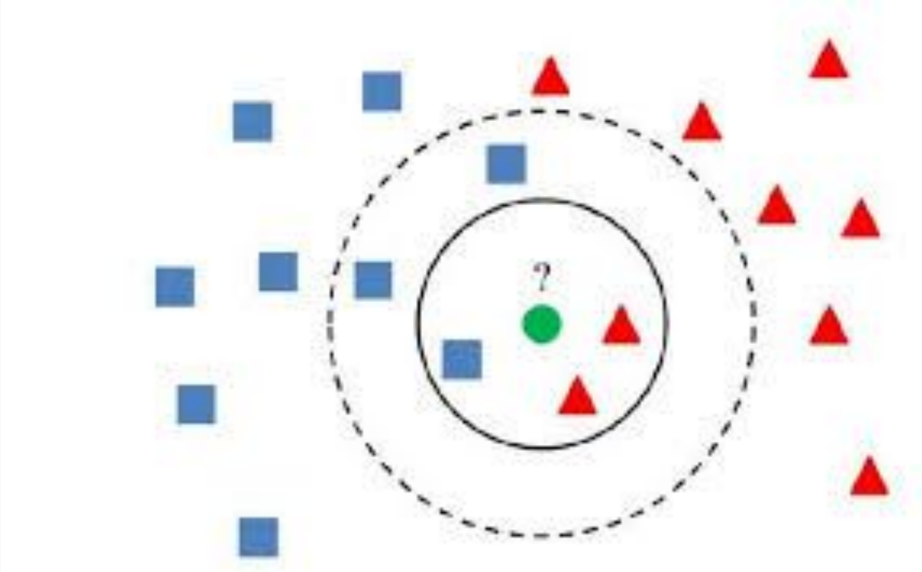
- It has 12 positive-positive and just 3 positive-negative
- It is the best study case

Conclusions

In this project, several machine learning classification models were developed to predict whether the first stage of a SpaceX Falcon 9 rocket would successfully land. The models tested included **Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)**. Hyperparameter tuning using GridSearchCV was applied to improve model performance.

After evaluating the models using cross-validation and testing on unseen data, **K-Nearest Neighbors (KNN)** was identified as the best-performing model for this dataset. The models achieved accuracies generally around **83–85%**, indicating that the selected features such as payload mass, orbit type, launch site, and booster version provide useful information for predicting landing success.

Overall, the analysis demonstrates that machine learning techniques can effectively predict Falcon 9 landing outcomes, which is valuable because successful booster landings significantly reduce the cost of space launches.



Appendix

```
#code to the accuracy bar chart
models = ['Logistic Regression', 'Random Forest', 'SVM', 'KNN']
accuracies = [accuracy_test, tree_score,svm_score,knn_score]
plt.figure(figsize=(10,6))
bars = plt.bar(models, accuracies, color='skyblue')
for bar, acc in zip(bars, accuracies):
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2.0, yval + 0.01, f'{acc:.2f}', ha='center', va='bottom')
plt.ylim(0, 1)
plt.title('Accuracy of Classification Models')
plt.ylabel('Accuracy')
plt.xlabel('Model')
plt.show()
```

Thank you!